Cairo University
Faculty of Engineering
Electronics & Electrical Communication Dept.



EECG325
3r year 2026 – 2027
Computer Dept.

Assignment 1: Random Processes

Objective

The objective of this assignment is to experiment with different types of random processes using Matlab

Overview

- A random process X(t) or X[n] is a collection of random variables indexed by time.
- Each experiment produces one realization (or sample function), but the process describes the behavior of all possible realizations.
- A process is strictly stationary if all statistical properties remain unchanged under time shifts.
- A process is Wide-Sense Stationarity (WSS) if:
  1) The mean is constant: $E[X(t)]$ = constant
  2) The autocorrelation depends only on the time lag |t1-t2| : $R_X(t1, t2) = R_X(t2 - t1)$
- WSS is widely used in engineering because it simplifies analysis and allows us to define the Power Spectral Density (PSD).

Assignment Part 1 — Identifying WSS vs non-WSS

Generate K = 100,000 realizations of white Gaussian noise $W_k[n] \sim \mathcal{N}(0,1)$, each of length N = 2000, and consider the two following processes

Process A: $X_A[k,n] = W_k[n]$
Process B: $X_B[k,n] = (1 + 0.5\,n)W_k[n]$

Tasks:

1) Plot 5 realizations of each process.
2) Compute the ensemble mean at n = 100, 1000, 2000.
3) Compute ensemble autocorrelation for (n1=100, n2=150) and (n1=1500, n2=1550)
4) Decide which process is WSS and which is not. Explain your answer

Assignment Part 2 — Fast and Slow WSS Processes

Generate K = 200 realizations of white Gaussian noise $W_k[n] \sim \mathcal{N}(0,1)$, each of length N = 1000, and consider the two following processes

Process A: $X_A[k,n] = W_k[n]$
Process B: $X_B[k,n] = \frac{1}{M}\sum_{i=0}^{M-1} W_k[n-i]$, $M = 10$

Tasks:

1) Plot one realization of each. Which process has fast variations and which has slow variations over time?
2) Compute the ensemble mean vs time.
3) Compute the ensemble autocorrelation R[τ] for τ = −100 to 100.
4) Compute PSD using FFT.
5) What is the relation between the ACF width, the PSD width, and the time variation. Explain your answer
6) Repeat steps 1-5 for M = 50. How does M affect the time variation, ACF width, and PSD width.